The Analyst Recommendation and Earnings Forecast Anomaly

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Equity Anomalies Extra Credit Project 17

Point-in-time IBES signals, CRSP returns, and a leakage-resistant backtest

Research Design Presentation

Cost-sensitivity results available; full performance audit in progress

Research question, mechanism, and hypotheses

Research question

Can public analyst recommendations predict *subsequent* stock returns after conservative information lags and trading costs?

Economic mechanism

- Analysts process firm, industry, and management information.
- Recommendation news may be incorporated gradually.
- Revisions may isolate new information better than recommendation levels.

Pre-specified hypotheses

1. Favorable consensus predicts higher next-month returns.
2. Upgrades outperform downgrades.
3. Signals strengthen when aligned with 12–2 momentum.
4. Effects may be larger for low-coverage firms.

Identification target

Return predictability must arise **after** the IBES information becomes observable—not from same-month returns, current identifiers, or missing outcome selection.

Data, historical links, and signal construction

Licensed data inputs

- **IBES Summary:** ticker, CUSIP, `statpers`, `meanrec`, `numrec`.
- **CRSP monthly:** PERMNO, return, delisting return, price, shares, share/exchange codes.
- **CRSP names:** historical CUSIP intervals for point-in-time linking.

Local Parquet cache

One-time WRDS download; files are excluded from Git and reused unless `-force` is supplied.

Consensus level

$$s^{C}_{i,t} = -\,\mathrm{meanrec}_{i,t}$$

Lower IBES scores are more favorable, so the sign is reversed.

Recommendation revision

$$s^{R}_{i,t} = \mathrm{meanrec}_{i,t-1} - \mathrm{meanrec}_{i,t}$$

Positive values are upgrades; negative values are downgrades.

Possible extensions (not in reported results): 12–2 mrec).

Point-in-time design and return alignment



Availability rule

Use the latest signal satisfying

$$\text{available_date}_i \leq \text{formation_date}_t.$$

A month-end snapshot cannot be traded at that same close.

Outcome rule

Weights formed in month t join only to calendar-month $t+1$ returns. CRSP outcomes combine

$$(1 + r_{i,t+1})(1 + \mathit{dlret}_{i,t+1}) - 1.$$

Portfolio construction and performance measurement

1. Keep CRSP common stocks on NYSE/AMEX/NASDAQ with price $\geq \$5$ and at least two recommendations.
2. Rank the selected signal within each formation month.
3. Equally weight the top decile long and bottom decile short (200% gross exposure).
4. Apply month-t weights to month-$t+1$ total returns.

Net strategy return

$$r_{p,t+1}^{net} = \sum_i w_{i,t} r_{i,t+1} - \frac{c}{10{,}000} \sum_i |w_{i,t} - w_{i,t-1}|,$$

with baseline $c = 10$ basis points.

Strategies used for reported results

Signal	Weight	Holding
Consensus	Equal	1 month
Revision	Equal	1 month

Exact implementation

Both legs rebalance monthly and are equally weighted. Recommendations enter after a one-day lag; formation-month signals earn the following month's return. Costs are charged per dollar traded.

Next steps for a better strategy

Consensus basket

- Average available analyst ratings with a minimum coverage rule.
- Select about 20 stocks with the strongest consensus.
- Form separate long and short baskets.
- Test roughly $500k per name and biweekly rebalancing.

Analyst-skill-weighted signal

- Use daily IBES recommendation events.
- Measure each analyst's volatility-normalized stock-minus-index return over 30 days after prior calls.
- Require at least 25 historical calls.
- Weight new recommendations by the analyst's historical score.

Selective trade overlay

- Keep only high-conviction broker/trader ideas.
- Combine them with the analyst signal and reject crowded exposures.
- Test a composite-score threshold above 0.75.
- Test about $100k per stock and a 10-day average holding period.

Portfolio construction

- Use a constrained optimizer such as Axioma.
- Control market, sector, style, liquidity, and concentration exposures.
- Compare Sharpe and total P&L across rebalance frequencies.

Scope boundary

These are proposed institutional extensions, not features of the reported backtest. The results in this deck use monthly IBES signals, equal-weighted top/bottom deciles, and next-month CRSP returns.

Leakage controls, automated tests, and data audit

Controls embedded in the pipeline

- Historical CUSIP validity; never current-ticker matching.
- Signal ranking does not inspect future-return availability.
- Missing held outcomes invalidate the portfolio month; never set to zero.
- Complete entries, exits, and rebalancing count toward turnover.
- Duplicate PERMNO/snapshot rows fail before revision construction.
- Extra CRSP outcome month cannot become a new formation month.

Seven automated tests currently pass

- Signal unavailable before publication date.
- Month-end snapshot receives the one-day lag.
- Future revisions cannot alter past signals.
- Month t maps to calendar month $t + 1$.
- Delisting returns compound correctly.
- Missing held outcome is not treated as zero.
- Revision uses only the prior snapshot.

Real-data gate

Before interpreting returns: audit link rate, ambiguous identifiers, duplicates, date coverage, delistings, outcome coverage, and portfolio counts.

Feasible extensions and required WRDS products

Extension	Additional design	Data	Feasibility
Momentum interaction	Double-sort revisions with trailing 12–2 returns	Existing CRSP	High
Coverage anomaly	Split signal efficacy by `numrec`	Existing IBES	High
Disagreement	Recommendation dispersion or analyst-detail cross-section	IBES optional/detail	Medium
Fundamentals	E/P, book-to-market, profitability, accruals	Compustat/ratios	Medium
Forecast revisions	Price-scaled changes in expected EPS	IBES estimates	Medium
PEAD interaction	Earnings surprise $\times$ recommendation revision	IBES actuals + CRSP	Medium–hard
Analyst skill	Condition on prior forecast/recommendation performance	IBES detail IDs	Hard
Broker quality	Point-in-time broker identity and classification	IBES detail + map	Hard

Recommended scope

Complete consensus and revision baselines first; then add momentum and coverage. Delay analyst/broker-level work because IBES identifiers have been reassigned historically.

Results so far: transaction costs materially weaken performance

Monthly baseline Sharpe by cost assumption

Cost	Consensus	Revision
0 bps	0.071	**0.513**
2 bps	0.059	0.373
5 bps	0.041	0.164
10 bps	0.011	−0.186

Consensus signal

Sharpe is low even before costs (0.071), but it decays relatively gradually to 0.011 at 10 bps.

Revision signal

Gross Sharpe is much stronger (0.513), yet the signal is highly cost-sensitive and becomes negative at 10 bps.

Economic implication

Any biweekly or 10-day institutional variant must earn enough incremental signal quality to offset its likely higher turnover. Selective thresholds, no-trade bands, and optimizer penalties should be tested.

Main conclusion

The monthly revision strategy dominates before costs, but becomes negative at 10 bps per dollar traded. The practitioner overlays are hypotheses for improving selectivity and risk control; the current Sharpe results do not yet validate them.

Current baseline evidence and planned robustness checks

Strategy	0 bps	2 bps	5 bps	10 bps
Consensus EW	0.071	0.059	0.041	0.011
Revision EW	0.513	0.373	0.164	−0.186

Entries are annualized Sharpe ratios under the stated per-dollar-traded cost assumption.

Additional outputs to estimate

- Monthly mean with HAC/Newey–West standard errors
- CAPM and multi-factor alpha
- Economic effect after transaction costs

Robustness

- Alternative lags, holding periods, and quantiles
- Size, price, liquidity, and coverage screens
- Subperiods and equal/value weighting

Interpretation rule

Do not claim an anomaly from one favorable specification. Emphasize stability, implementation costs, factor adjustment, and the number of alternatives tested.

Limitations, execution plan, and primary sources

Known limitations

- WRDS schemas and entitlements differ across institutions.
- IBES analyst/broker identifiers have changed over time.
- Exact intraday publication times are generally unknown.
- Equal weighting can exaggerate small-stock implementability.
- Multiple specifications create data-mining risk.

Primary sources

- WRDS I/B/E/S Overview and ICLINK documentation
- WRDS CRSP Monthly Stock documentation
- *Handbook of Equity Market Anomalies*, Chapter 3

Execution plan

1. Complete one-time WRDS Parquet cache.
2. Audit schemas, links, dates, duplicates, and outcomes.
3. Complete all metrics for the two reported EW baselines.
4. Add factor regressions and robustness specifications.
5. Test consensus-basket and analyst-skill variants separately.
6. Compare monthly, biweekly, and 10-day designs after costs.